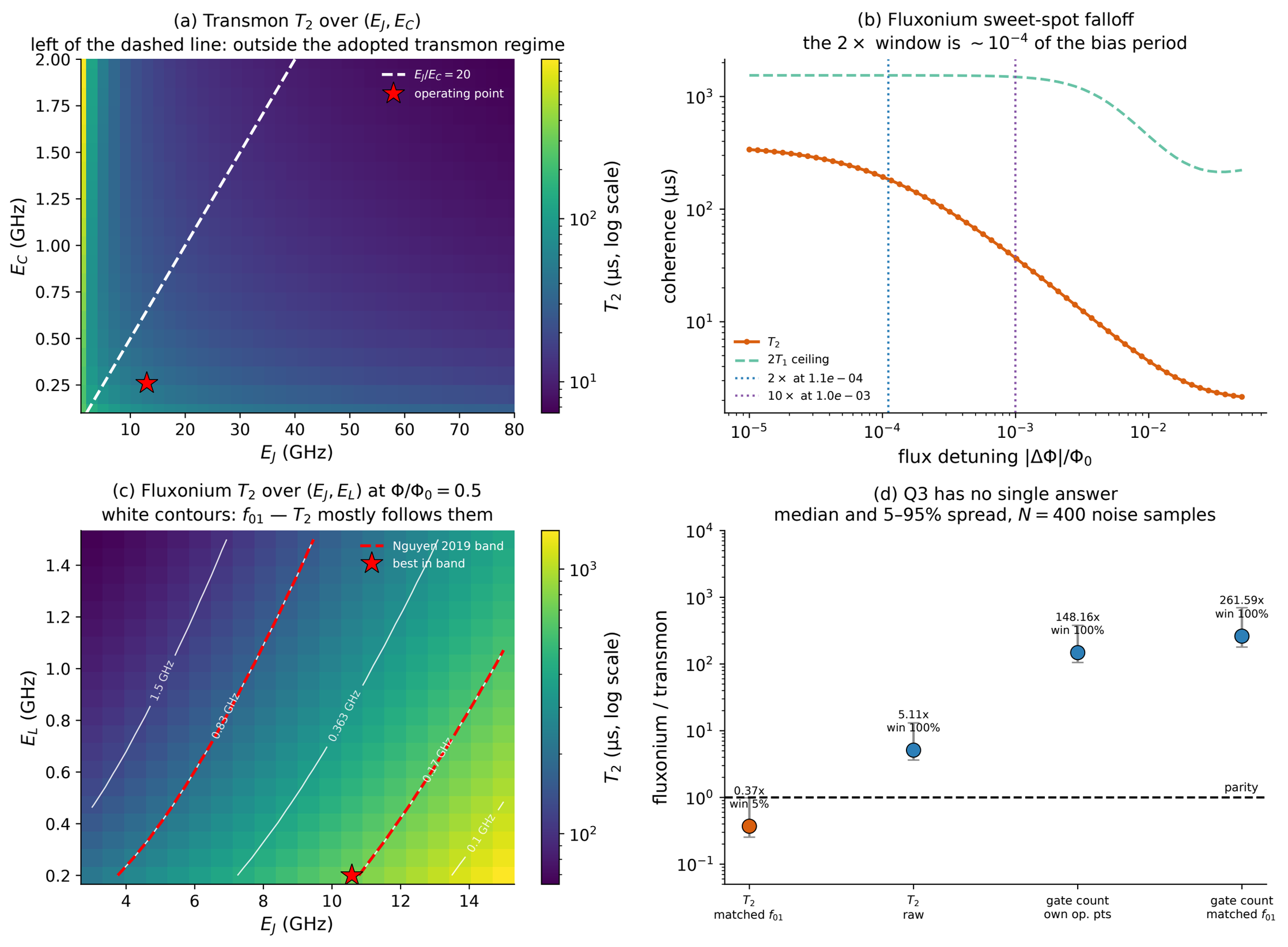

(a) Transmon T_2 over (E_J, E_C)
left of the dashed line: outside the adopted transmon regime
$E_J/E_C = 20$
operating point
E_C (GHz)
E_J (GHz)
T_2 (µs, log scale)
(b) Fluxonium sweet-spot falloff
the 2 × window is $\sim 10^{-4}$ of the bias period
coherence (µs)
T_2
$2T_1$ ceiling
2 × at 1.1e − 04
10 × at 1.0e − 03
flux detuning $|\Delta\Phi|/\Phi_0$
(c) Fluxonium T_2 over (E_J, E_L) at $\Phi/\Phi_0 = 0.5$
white contours: f_{01} — T_2 mostly follows them
Nguyen 2019 band
best in band
1.5 GHz
0.83 GHz
0.363 GHz
0.17 GHz
0.1 GHz
E_L (GHz)
E_J (GHz)
T_2 (µs, log scale)
(d) Q3 has no single answer
median and 5–95% spread, $N = 400$ noise samples
fluxonium / transmon
0.37x
win 5%
5.11x
win 100%
148.16x
win 100%
261.59x
win 100%
parity
T_2 matched f_{01}
T_2 raw
gate count own op. pts
gate count matched f_{01}